

Supplementary to Homework 7 QueryServlet with Sample Database Testing Class

Knowledge requirement: MySQL Server, JDBC connection, SQL Statement and JSP
File: user-query.jsp, show-result.jsp and QueryServlet.java only

Command to start MySQL Server (Database Server storing multiple databases; Each database stores multiple tables)

```
mysqld --initialize-insecure
```

```
mysqld
```

(Keep this command-prompt opened at the background)

How to connect to the MySQL server that is running on the local machine (localhost) using the username 'root' created by default

```
mysql -h localhost -u root
```

--Create a new database named Books

--Select the Books database

--Create table named Authors under the Books database

```
CREATE DATABASE IF NOT EXISTS Books;
```

```
USE Books;
```

-- Create the table, Authors with three different field. No need to insert value to the first field as it will create in sequence after insertion of each entry. CREATE TABLE

```
Authors (    authorID INT AUTO_INCREMENT PRIMARY KEY,
```

```
    FirstName VARCHAR(255) NOT NULL,
```

```
    LastName VARCHAR(255) NOT NULL
```

```
);
```

-- Create authors; Insert 4 records; All record will be stored permanently under C:\Temp\EIE3320\MySQL\mysql-8.2.0-winx64\data

```
INSERT INTO Authors (FirstName, LastName) VALUES
```

```
    ('Harvey', 'Deitel'),
```

```
    ('Paul', 'Deitel'),
```

```
    ('Andrew', 'Goldberg'),
```

```
    ('David', 'Coffnes');
```

-- If you wish to add one more record, for example, please login again if you have not and type the SQL statement to insert one more record:

```
mysql -h localhost -u root
```

```
USE Books;
```

```
INSERT INTO Authors (FirstName, LastName) VALUES ('Peter', 'Chan');
```

-- Check whether you have successfully created all new records; SQL statement should be ended with semi colon

```
mysql> select * from authors;
```

AuthorID	FirstName	LastName
1	Harvey	Deitel
2	Paul	Deitel
3	Andrew	Goldberg
4	David	Coffnes

```
4 rows in set (0.00 sec)
```

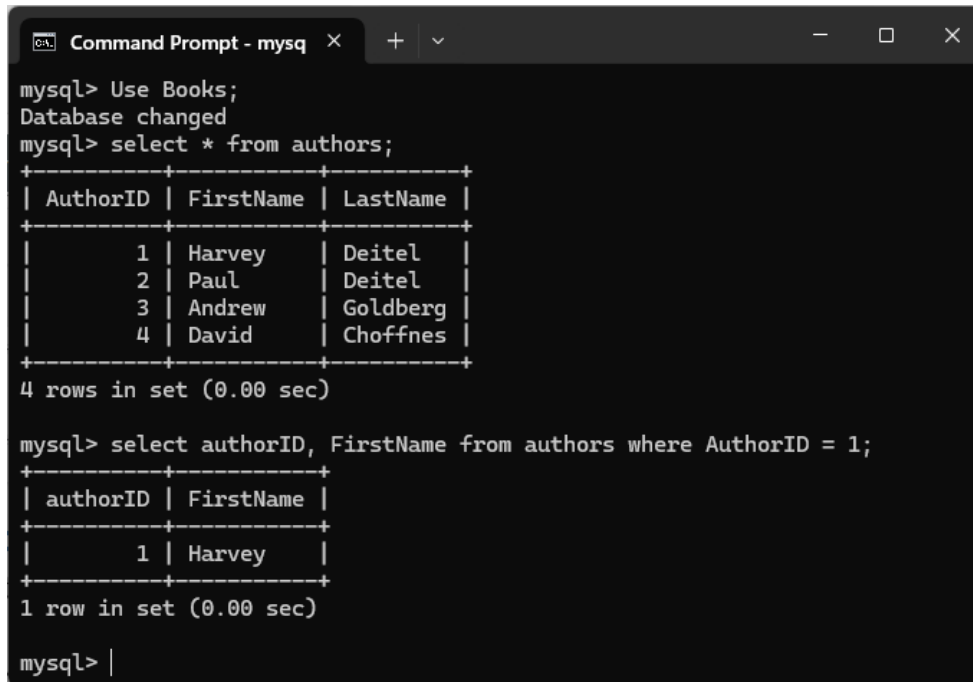
Sample Java class to test the database connection

Assume your MySQL server is running locally on port 3306, with the username "guest" and password "guest," and the "Books" database has been created from Books.sql. Additionally, ensure you have copied the mysql-connector-j-8.4.0.jar file to the tomcat-11.x.x\lib folder.

In Command Prompt, `c:\Temp\EIE3320\MySQL\mysql-8.4.3-win64\bin>mysql`

You may open another command to test your database or display partial information:

For example, type `select authorID, FirstName from authors where AuthorID = 1;`

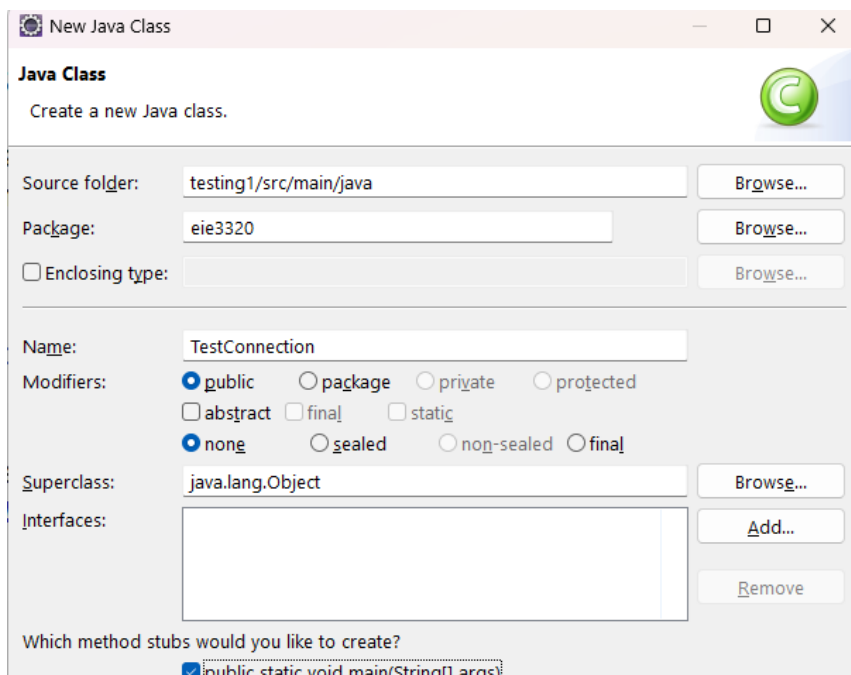


```
mysql> Use Books;
Database changed
mysql> select * from authors;
+-----+-----+-----+
| AuthorID | FirstName | LastName |
+-----+-----+-----+
| 1 | Harvey | Deitel |
| 2 | Paul | Deitel |
| 3 | Andrew | Goldberg |
| 4 | David | Choffnes |
+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> select authorID, FirstName from authors where AuthorID = 1;
+-----+-----+
| authorID | FirstName |
+-----+-----+
| 1 | Harvey |
+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

Right click over the project name, choose New > Class.



New Java Class

Create a new Java class.

Source folder:

Package:

☐ Enclosing type:

Name:

Modifiers: ☒ public ☐ package ☐ private ☐ protected
☐ abstract ☐ final ☐ static
☒ none ☐ sealed ☐ non-sealed ☐ final

Superclass:

Interfaces:

Which method stubs would you like to create?
☒ `public static void main(String[] args)`

```

package eie3320;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class TestConnection {

    public static void main(String[] args) {

        String jdbcUrl = "jdbc:mysql://localhost:3306/Books";

        String user = "guest";

        String password = "guest";

        try {

            // Load MySQL connector

            Class.forName("com.mysql.cj.jdbc.Driver");

            // Create database connection

            Connection connection = DriverManager.getConnection(jdbcUrl, user, password);

            System.out.println("Database connection successful!");

            // Close connection

            connection.close();

        } catch (ClassNotFoundException e) {

            System.err.println("MySQL JDBC driver not found.");

            e.printStackTrace();

        } catch (SQLException e) {

            System.err.println("Error connecting to the database.");

            e.printStackTrace();

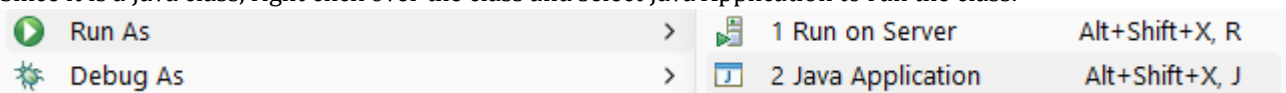
        }

    }

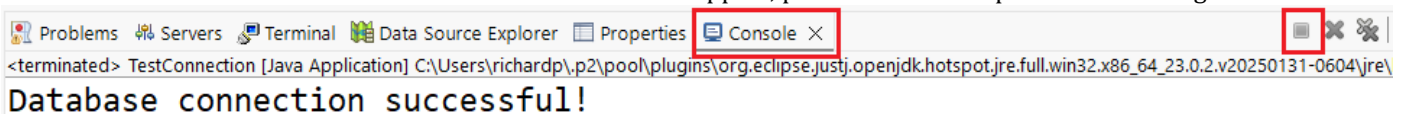
}

```

Since it is a Java class, right click over the class and select Java Application to run the class.



The result will be shown in the console. If the result does not appear, please click the stop button on the right first.



--You may try the following SQL to display partial information:

```
select authorID, FirstName from authors where AuthorID = 1;
```

//Implement doGet or doPost method in your Servlet which extends HttpServlet

//Get author id from the input textbox

//Create attribute on the request object and

//store the value as a pair of key and value format

```
String authorID = request.getParameter("authorID");
```

```
request.setAttribute(_____, _____);
```

//Connect the db try

```
{
```

```
    Class.forName("com.mysql.cj.jdbc.Driver");
```

```
    String jdbcUrl = "jdbc:mysql://localhost:3306/Books";
```

```
    String user = "guest";
```

```
    String password = "guest";
```

```
    Connection connection = DriverManager.getConnection(jdbcUrl, user, password);
```

//Query the db

//The ? is a placeholder for the value that we want to compare authorID to

```
String query = "SELECT FirstName, LastName FROM authors WHERE authorID = ?";
```

```
PreparedStatement preparedStatement = connection.prepareStatement(query);
```

//This line is setting the value of the placeholder ? in the SQL query.

//The first parameter 1 indicates that we're setting the first placeholder in the SQL query.

//The second parameter authorID is the value that we're setting the placeholder to

```
preparedStatement.setString(1, _____ );
```

//This code is executing a SQL query that selects the FirstName and LastName from the authors table for a specific authorID, and then storing the results of

//that query in a ResultSet object.

//e.g. "FirstName" refers to the table name while

//resultSet.next is to read the first line of result

```
ResultSet resultSet = preparedStatement.executeQuery();
```

```
    if (resultSet.next())
```

```
{
```

```
    String firstName = resultSet.getString("FirstName");
```

```
    ...
```

```
    request.setAttribute(_____, _____);
```

```
} else {
```



```

        request.setAttribute("error", "Author not
found.");    }

        //Close connection
resultSet.close();
preparedStatement.close();
connection.close();

    } catch (ClassNotFoundException | SQLException e)
{
        request.setAttribute("error", "Error connecting to the
database.");

        e.printStackTrace();
    }

//Send to show-result.jsp;
//Forward the client's request from a servlet to a JSP page
//which will generate the response.

    request.getRequestDispatcher("/_____").forward(request, response);
}

e.g. In the, you may extract the attribute of the HTTP request object through:
request.getAttribute(_____);

```

1. Input author ID for searching from the database



```

mysql> select * from Authors;
+-----+-----+-----+
| authorID | FirstName | LastName |
+-----+-----+-----+
| 1 | Harvey | Deitel |
| 2 | Paul | Deitel |
| 3 | Andrew | Goldberg |
| 4 | David | Coffnes |
| 5 | Peter | Chan |
+-----+-----+-----+
5 rows in set (0.00 sec)

```

2. Forward the request from a servlet to a JSP page to generate the response (User will not browse show-result.jsp directly.)



Homework 7 ATM

Knowledge requirement: html and servlet

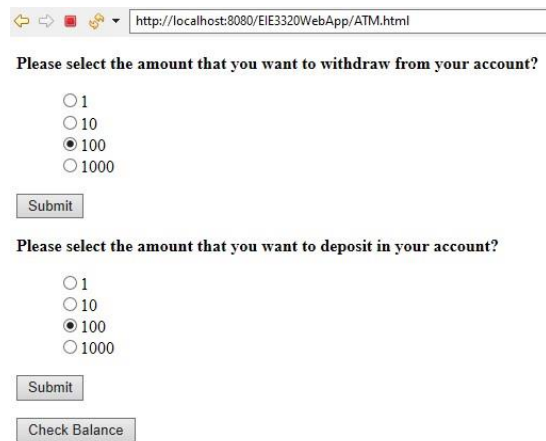
File: ATM.html, CheckBlanaceServlet.java, CreditServlet.java and DebitServlet.java only

Assume the initial balance is zero. No database needed. Use HttpSession to store balance. e.g. if balance is null, set balance to zero.

Please develop your one html page and three servlet classes only:

ATM.html, CheckBlanaceServlet.java, CreditServlet.java and DebitServlet.java

1. Homepage



The screenshot shows the homepage of the ATM application. The browser address bar displays 'http://localhost:8080/EIE3320WebApp/ATM.html'. The main content area contains two sections. The first section is titled 'Please select the amount that you want to withdraw from your account?' and features four radio button options: 1, 10, 100 (which is selected), and 1000. Below these options is a 'Submit' button. The second section is titled 'Please select the amount that you want to deposit in your account?' and also features four radio button options: 1, 10, 100 (which is selected), and 1000. Below these options is a 'Submit' button. At the bottom of the form is a 'Check Balance' button.

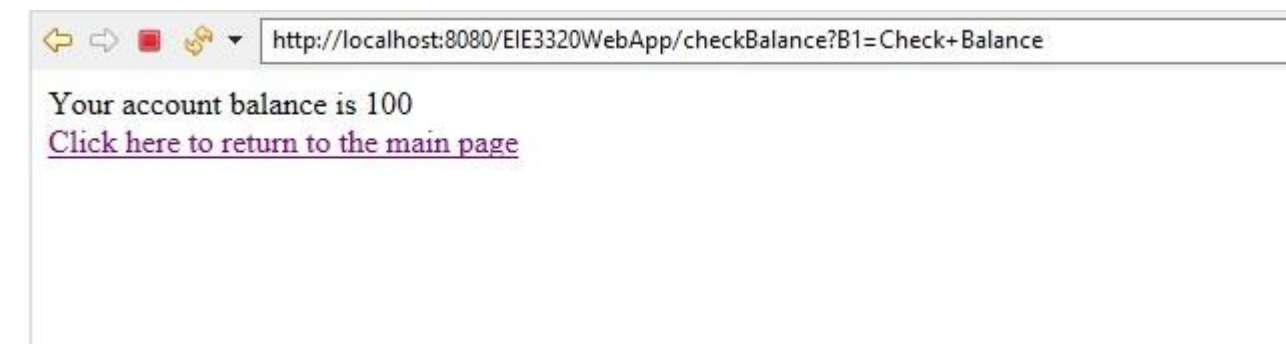
2. After clicking Check Balance



The screenshot shows the ATM page after clicking the 'Check Balance' button. The browser address bar displays 'http://localhost:8080/EIE3320WebApp/checkBalance?B1=Check+Balance'. The main content area displays 'Your account balance is 0' in a large font. Below this text is a purple hyperlink that reads 'Click here to return to the main page'.

3. Click the hyperlink to return back to homepage

4. Click the second Submit button to deposit 100 dollars. 5. Click Check balance



The screenshot shows the ATM page after clicking the 'Check Balance' button. The browser address bar displays 'http://localhost:8080/EIE3320WebApp/checkBalance?B1=Check+Balance'. The main content area displays 'Your account balance is 100' in a large font. Below this text is a purple hyperlink that reads 'Click here to return to the main page'.

6. Click the hyperlink to return to homepage

7. Select the first 10 and click the first Submit button to withdraw 10 dollars.

8. Click the Check Balance



The screenshot shows the ATM page after clicking the 'Check Balance' button. The browser address bar displays 'http://localhost:8080/EIE3320WebApp/checkBalance?B1=Check+Balance'. The main content area displays 'Your account balance is 90' in a large font. Below this text is a purple hyperlink that reads 'Click here to return to the main page'.

Hints:

1. Please use `Integer.parseInt()` to convert all value from Input textbox into integer for calculation.
2. Please use `response.sendRedirect("ATM.html");` to make a request or redirect to a URL/webpage and the current servlet will be stopped.